STL Containers

Containers hold data. Memory layout, operation costs and the choice between vector, array, deque and list are the parts that matter most. Assumes Pointers and References.

The big picture



Algorithms never touch containers, only iterator ranges.

One sort works on any container that offers the right iterators.

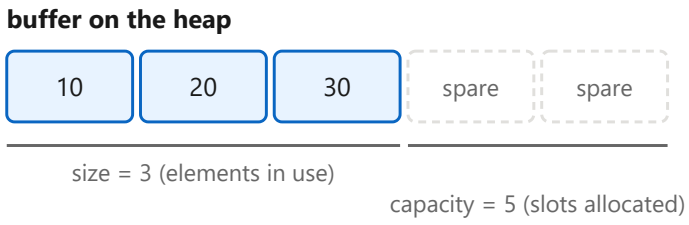
PART	WHAT IT IS	EXAMPLES
Containers	classes that own and organise a collection of elements	`vector`, `map`, `set`, `deque`
Iterators	generalised pointers that walk a container	`v.begin()`, `m.find(k)`
Algorithms	generic functions that operate on iterator *ranges*	`sort`, `find`, `accumulate`

Algorithms see only `[first, last)`, so one `std::sort` serves a vector, array or deque. That separation is the design.

```
#include <vector>       // std::vector
#include <map>          // std::map, std::multimap
#include <unordered_map>// std::unordered_map
#include <set>          // std::set
#include <algorithm>    // std::sort, std::find, std::transform ...
#include <numeric>      // std::accumulate, std::iota
#include <string>       // std::string
```

DEFAULT CHOICE `std::vector`. Contiguous memory is so cache-friendly it often beats "better" containers.

`vector`: the workhorse



push_back, buffer full:
allocate bigger, move all, free old
= amortised O(1), but every iterator and pointer into it is invalidated

reserve() grows capacity only; resize() changes size.

A `vector` is one contiguous heap array with a **size** and a **capacity**. When full, `push_back` allocates a bigger buffer, moves everything and frees the old one: occasional O(n), **amortised O(1)**.

```
std::vector<int> v;
// capacity 100, size still 0: no reallocation for the next 100 push_backs
v.reserve(100);
v.push_back(1);          // copy/move an existing object in
v.emplace_back(2);       // construct the element in place: no temporary
// size becomes 10; new elements are value-initialised (0)
v.resize(10);

// no bounds check: out-of-range is undefined behavior
v[3];
v.at(3);                 // bounds-checked: throws std::out_of_range
v.front(); v.back(); v.data();   // data() is the raw int* for C APIs
v.size(); v.capacity(); v.empty();
```

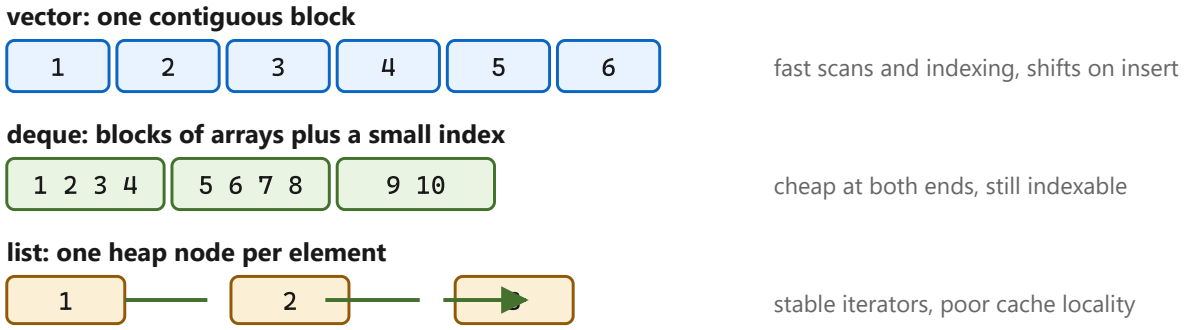
RESERVE VS RESIZE `reserve(n)` only allocates; `resize(n)` creates elements. `v[5] = x` after only `reserve(10)` is UB.

BRACES VS PARENTHESES

```
// parentheses: size 3, every element 7 → {7, 7, 7}
std::vector<int> a(3, 7);
// braces: the initializer_list constructor wins → {3, 7}
std::vector<int> b{3, 7};
```

With braces, a constructor taking `std::initializer_list` is preferred whenever it can match.

Other sequence containers



Memory layout explains the costs: contiguous is cache-friendly, linked is flexible but scattered.

CONTAINER	STORAGE	BEST AT	WATCH OUT
`std::array<T,N>`	fixed size, inline (stack)	size known at compile time; zero overhead	cannot grow
`std::vector<T>`	one contiguous heap block	almost everything; fast iteration and indexing	insert/erase in the middle is O(n)
`std::deque<T>`	chunks of arrays	fast push/pop at **both** ends, still indexable	not contiguous
`std::list<T>`	doubly linked nodes	O(1) insert/erase anywhere *if you hold an iterator*; iterators stay valid	no indexing, a heap allocation per element, poor cache behaviour
`std::forward_list<T>`	singly linked nodes	minimal memory list	forward iteration only

OPERATION	VECTOR	DEQUE	LIST
index `[i]`	O(1)	O(1)	not available
`push_back`	amortised O(1)	O(1)	O(1)
`push_front`	O(n)	O(1)	O(1)
insert / erase in the middle	O(n)	O(n)	O(1) at a known position
find a value	O(n)	O(n)	O(n)

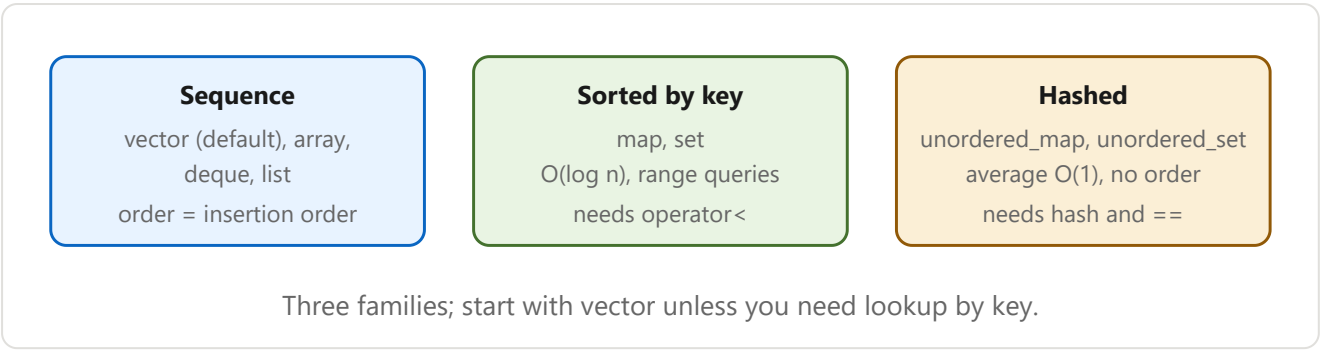
Container adaptors

They wrap another container and restrict the interface: `std::stack` (LIFO: `push/pop/top`), `std::queue` (FIFO: `push/pop/front`) and `std::priority_queue` (a heap: `top()` is the **largest** element by default).

```
// largest first
std::priority_queue<int> maxHeap;
// smallest first
std::priority_queue<int, std::vector<int>, std::greater<int>> minHeap;
```

Which container should I use?

I NEED TO...	USE	WHY
iterate over items, access by index, mostly append at the end	`std::vector`	Contiguous memory: fastest iteration and indexing, amortised O(1) `push_back`. This is the default choice.
hold a fixed number of items known at compile time	`std::array`	Lives inline (no heap), knows its size, zero overhead over a C array.
push and pop at both ends (a deque of work)	`std::deque`	O(1) push/pop at front and back while still allowing indexing.
insert/erase often in the middle while holding iterators that must stay valid	`std::list`	O(1) splice/insert/erase at a known position and stable iterators. Costs: a heap allocation per node and poor cache locality. Measure before choosing it.
look up values by key; order does not matter	`std::unordered_map`	Hash table: average O(1) find/insert/erase. Needs a hash and `operator==`.
look up by key AND iterate in sorted order, or find the nearest key	`std::map`	Balanced tree: O(log n), sorted iteration, `lower_bound` / `upper_bound` for range queries.
check membership / remove duplicates (no values attached)	`std::unordered_set`	Hash set: average O(1) contains. Use `std::set` if you also need the elements sorted.
count how often each word occurs	`std::unordered_map<std::string, int>`	`counts[word]++` inserts a zero on first sight and increments in one expression.
last-in first-out (undo, DFS, matching brackets)	`std::stack`	Adaptor over deque/vector exposing only push, pop and top.
first-in first-out (task queue, BFS)	`std::queue`	Adaptor over deque exposing push, pop, front and back.
always process the largest (or smallest) item next	`std::priority_queue`	A binary heap: O(log n) push/pop, O(1) top. Largest first by default; pass `std::greater` for smallest first.



Three families; start with vector unless you need lookup by key.

KEY POINT "vector vs list?" Answer with *memory*: contiguous means cache-friendly, so vector wins even for middle inserts at modest sizes. List wins for stable iterators and cheap splicing.

Common pitfalls

`optional` and `variant` are covered in optional, variant and expected.

PITFALL	FIX
Using an iterator/pointer/reference after a `push_back` reallocated the vector	`reserve` up front, or re-fetch the iterator, or index instead
`map[key]` silently inserts	use `find` / `contains` to test; `at()` to read-or-throw
`v.size() - 1` on an empty vector (`size()` is unsigned, so it wraps to a huge number)	check `empty()` first or write `i + 1 < v.size()`
Passing a big container by value	pass `const std::vector<T>&`
`std::vector<bool>` is a packed-bit special case with proxy references	use `std::vector<char>` or `std::bitset`
`auto x = v[0]` copies the element	`const auto& x = v[0]`
`std::find` on a `map` is O(n)	use the member `m.find(key)`, O(log n)

Further reading by section

Where each section of this lesson is explained in more depth. Pages are PDF viewer pages (the number your PDF reader shows).

SECTION	BOOK	PART OF THE BOOK	PDF PAGES
vector: the workhorse	C++ Primer	3.3 Library vector type	125 to 134
	C++ Primer	9.3 Sequential container operations	370 to 383
	C++ Primer	9.4 How a vector grows	384 to 388
	Programming: Principles and Practice	3.5 to 3.6 Functions; vector introduced	81 to 89
	Programming: Principles and Practice	17.6 to 17.9 Other operations, remaining Vector problems, changing size	513 to 523
	Effective Modern C++	Item 42, Consider emplacement instead of insertion	310 to 320
	C++ Move Semantics	Chapter 15 Move semantics in standard library types	229 to 250
Other sequence containers	C++ Primer	9.1 to 9.2 Overview and the container library	355 to 369
	C++ Primer	9.6 Container adaptors	397 to 402
	C++ Primer	10.6 Container-specific algorithms	444 to 447
	Programming: Principles and Practice	19.1 to 19.3, 19.6 Containers, sequences and iterators, linked lists, vector/list/string	559 to 572 and 585 to 586
Which container should I use?	Programming: Principles and Practice	20.1 to 20.6 map, unordered_map, set, container overview	591 to 609
Common pitfalls	C++ Primer	9.3 Sequential container operations	370 to 383